



An assembly process planning pipeline for industrial electronic equipment based on knowledge graph with bidirectional extracted knowledge from historical process documents

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Abstract

Assembly is an essential stage in industrial electronic equipment manufacturing and needs to meet the complexity of manufacturing. Therefore, the assembly process planning for industrial electronic equipment still relies on the experiences of planners. The advent of knowledge graphs brings an opportunity to achieve automated assembly process planning. Thus, extracting process knowledge from historical assembly process documents and constructing assembly process knowledge graphs are indispensable. However, the complexity of industrial electronic equipment manufacturing leads to assembly process documents containing more complex assembly relations, longer texts, and high-density assembly entities. These characteristics pose challenges to assembly process knowledge extraction and knowledge graph modeling. The confidentiality of assembly process documents further hinders the development of this field. To address these challenges, we propose a pipeline for achieving assembly process planning from historical assembly process documents. First, we construct an assembly process dataset using historical assembly process documents from an industrial electronic equipment enterprise. Then, we propose a global relation-driven bidirectional extraction model, which automatically constructs the assembly process knowledge graph. In addition, we also propose a knowledge graph-based matching and searching method to support process planning. The proposed model is evaluated on the constructed dataset and a publicly accessible equipment fault diagnostic dataset, achieving F1-scores of 92.9% and 87.9%, respectively. Experimental results demonstrate that the proposed model achieves state-of-the-art performance on both datasets. Furthermore, we construct an assembly process knowledge graph for industrial electronic equipment and perform assembly process planning, which validates the feasibility of our pipeline.

Keywords Assembly process knowledge graph · Process knowledge extraction · Assembly process knowledge · Assembly process planning

Introduction

Industrial electronic equipment has complex structures and various components, and the assembly process needs to satisfy the multiple varieties, small-batch, and flexible recon-

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figuration characteristics (Yin et al., 2022). This requires that industrial electronic equipment assembly process planning meet product characteristics and respond quickly to dynamic product iterations. Industrial electronic equipment enterprises have accumulated numerous historical assembly process documents in structured, semi-structured, and unstructured formats (Wang, Zheng, Lv, Bao and Zhang, 2019). Currently, most manufacturing enterprises manage historical assembly process documents with traditional computer-aided process planning (CAPP) systems (Kutin, Dolgov, Sedykh and Ivashin, 2018). Designers use traditional CAPP systems for assembly process planning, which relies on empirical knowledge and historical assembly process documents. This approach is a manual or semi-automated assembly process planning, which only partially improves efficiency (Zhang, 2022). Traditional CAPP systems can handle assembly process planning for common industrial equipment. However, for complex products such as industrial electronic equipment, the assembly process planning must be fast, dynamic, and flexible according to production requirement variations. With the advent of the intelligent manufacturing era, mining and reusing assembly process knowledge from historical assembly process documents has become a significant trend in the assembly process planning for complex industrial electronic equipment (Zhou, Bao, Liu and Song, 2020). Transform historical assembly process documents into structured process knowledge, construct assembly process knowledge graph (APKG), then develop the intelligent industrial electronic equipment process planning pipeline driven by APKG, which can flexibly tackle multi-species mixed-flow assembly line, small batch, and flexible assembly line reconfiguration of the assembly procedure. Knowledge Graph (KG) is a large-scale semantic network for knowledge representation, originally proposed by Google in 2012 and successfully applied in search engines (Singhal et al., 2012). KG is widely used in finance, healthcare, education, manufacturing, and engineering (Abu-Salih, 2021). In the field of equipment manufacturing, KG also plays an important role in diagnosis (Fan, Xu, Jiang and Ding, 2022), risk prediction (Kurniawan, Ekelhart, Kiesling, Quirschmayr and Tjoa, 2022), and production resource allocation (Zhou, Bao et al., 2021). KG uses large-scale semantic networks to delineate entities and interrelations in the objective world (Hogan et al., 2021). These are represented and stored as knowledge triple (subject, relation, object) or (s, r, o) . The subject denotes the subject entity, the object denotes the object entity, and the subjects and objects are the nodes in KG. The relation denotes the relationship between the subject and the object, and the relations are the edges of KG.

KG has the following significant advantages over traditional knowledge engineering (e.g., knowledge-based expert systems (Ritchi, Turban and Aronson, 2011)): discovery of implicit and procedure knowledge, self-learning, com-

pleteness, and consistency (Chen, Jia and Xiang, 2020). The analysis reveals that the assembly process planning of industrial electronic equipment has the following characteristics and research challenges: (1) Industrial electronic equipment has more complicated structure and numerous components than common mechanical equipment, which requires the assembly process planning to address more complicated assembly sequences and materials; (2) The assembly of industrial electronic equipment has the characteristics of multi-species mixed-flow assembly line, small batch, and flexible assembly line reconfiguration (Shen, Li, Zhou, Jiang and Bao, 2023). Thus, the assembly process planning must have the dynamic iterative ability to quickly respond to production variations; (3) The assembly of industrial electronic equipment requires high precision and high reliability, which requires the APP to satisfy. To overcome the above challenges, KG is utilized to realize assembly process planning. KG is well suited to represent the process knowledge generated during manufacturing, forming the assembly process knowledge graph (APKG). The APKG can self-learn, discover implicit process knowledge, update process knowledge, and infer new assembly processes (Zhou et al., 2021; Guo, Yan, Li, Yang and Lu, 2022). Therefore, the rapid and automatic construction of the APKG is critical to achieving intelligent manufacturing for enterprises. Extracting process knowledge triple (s, r, o) from historical assembly process data is essential in constructing a large-scale APKG.

Most manufacturing enterprises employ computer-aided process planning (CAPP) systems (Xu, Wang and Newman, 2011) for assembly process planning and have accumulated large amounts of historical assembly process data, including structured, semi-structured, and unstructured data. The structured and semi-structured assembly process data mostly consists of drawings and tables. For instance, STEP-compliant CAD files, these data are prone to extract assembly process knowledge (Xu, Wang and Newman, 2011). However, most assembly process data exist in unstructured natural language text documents, including process flow documents, engineering documents, operational documents, and normalized standards documents (Liu et al., 2022). These documents describe the man, machine, material, method, measurement, and environment of process planning, but the documents are susceptible to the experience and competence of the process planners. The elements of process planning are implicit in these documents, which are characterized by unstructured, multiple data sources and low semantic associations. Hence, extracting assembly process knowledge triples from unstructured process documents is essential in constructing the APKG (Guan, Du and Yang, 2022). APKG can support the process planners in effectively organizing and reusing the assembly process knowledge and reduce the difficulty of process planning.

In comparison to the open world, the modeling of the APKG has the following challenges: (1) the documents are longer than the public dataset; (2) the quantity of entities is more than the public dataset, which increases the difficulty of entity recognition; (3) more complex relation semantics between entities; and (4) There are no knowledge triple datasets for assembly processes, which significantly impedes studying APKG modeling methods. To tackle these challenges, we first construct an assembly process dataset about industrial electronic equipment. The process data are sourced from an industrial electronic equipment enterprise. The dataset contains 9570 assembly process documents, 11,179 assembly entities, and 13 assembly relation categories. Then, we propose a global relation-driven bidirectional extraction model for extracting assembly process knowledge triples to enable the APKG modeling for long texts, high-density entities, and complex semantics assembly process documents. In addition, we also propose a knowledge graph-based matching and searching (KGMS) method to support process planning. Finally, we construct an industrial electronic equipment APKG and achieve assembly process planning based on the APKG. The above procedures compose a complete pipeline for constructing the APKG from historical assembly process documents and achieving assembly process planning.

In summary, the main contributions of this paper are as follows:

- (1) We propose a global relation-driven bidirectional extraction (GRBE) model to extract assembly process knowledge triples with long documents, high-density entities, and complex relation semantics. The proposed model can automatically construct the APKG.
- (2) The GRBE model demonstrates state-of-the-art performance on the assembly process dataset.
- (3) We construct an APKG based on the GRBE model and propose a knowledge graph-based matching and searching (KGMS) method to support process planning on the APKG. Moreover, we quantitatively evaluate the generated assembly process planning paths using the Jaccard index.

The remainder of this paper is organized as follows. Section 2 summarizes the related works. Section 3 focuses on our proposed pipeline. Section 4 shows the experimental results. Section 5 presents the conclusion and future work.

Related works

Currently, KG modeling methods in the process planning can be classified into ontology-based methods, feature-based methods, pipeline-based methods, and joint extraction-based methods, as shown in Table 1.

Ontology-based methods. These methods rely on ontologies and rules designed by experts and are applied to knowledge-based expert systems. Many classical methods have achieved excellent results in regular process knowledge extraction and regular APKG. Kang et al. (2015) utilized the OWL ontology for defining regularized manufacturing knowledge from unstructured manufacturing manual documents, thereby forming rule-based knowledge triples. Yang et al. (2018) employed n-gram for preprocessing digitized pre-manufacturing process documents and constructed domain ontology to transform the documents into structured data and extract the pre-manufacturing process knowledge in the automobile manufacturing domain. Kang et al. (2019) tackled the issue of ambiguity in manufacturing documents by employing ontology to obtain complex manufacturing knowledge from unstructured sources. Du et al. (2022) proposed an ontology construction method that automatically extracts case entity concepts and related attributes from the case assembly process text and fuses the assembly domain ontology template to generate an assembly case ontology. Bao et al. (2022) proposed an ontology-based method for assembly process modeling, addressing the challenges of non-standardized descriptions and insufficient semantics in knowledge extraction for assembly processes. Kang et al. (2021) investigated the rule extraction problem for unstructured English texts in the manufacturing domain and proposed a formal rule extraction framework. With the above methods, the ontology or rule-based methods achieved considerable success. However, manual experiences suffer from unstable reliability and high cost. With weak generalization, these methods are difficult to adapt to complex linguistic texts. It is difficult for these methods to discover the deep semantic knowledge implicit in the documents, which is crucial for the assembly process knowledge extraction and APKG modeling.

Feature-based methods. These methods employ manually designed process knowledge features or learn process knowledge features using machine learning algorithms. For example, Leng and Jiang (2016) proposed a deep learning method based on stacked denoising autoencoders for manufacturing relation extraction, which was more applicable than manually designed features. Weston et al. (2019) proposed a similar method for large-scale named entity recognition from the material science literature libraries. Krahe et al. (2021) established the knowledge of CAD models based on CAD model tree design features. Owing to the distinct characteristics of the CAD model tree, features that are manually designed exhibit greater accuracy and are more cost-effective than features derived through learning.

Pipeline-based methods. These methods divide process knowledge triple extraction into two tasks: process named entity recognition and process relation classification. The

Table 1 Comparison of KG modeling methods

Type	Method Descriptions	Characteristics
Ontology-based methods	These methods rely on ontologies and rules designed by experts and are applied to knowledge-based expert systems for CAPP	1) Unstructured and complex process knowledge representation; 2) Simple logical inference; 3) Complex KG modeling capability
Feature-based methods	These methods employ manually designed process knowledge features or learn process knowledge features using machine learning algorithms	1) Manually designed process knowledge feature extractor; 2) Logical inference; 3) KG modeling relies on manual experiences
Pipeline-based methods	These methods divide process knowledge triple extraction into two tasks: process named entity recognition and process relation classification	1) Automated learned process knowledge feature extractor; 2) Powerful logical inference; 3) KG modeling requires two steps
Joint extraction-based methods	These methods can simultaneously extract process entities and process relations	1) Automated learned process knowledge feature extractor; 2) Powerful logical inference; 3) End-to-end KG modeling in one step

process entity pairs are first extracted, and then the process entity pairs are categorized to determine their relations. These methods have the advantage of simplicity but break up the semantic connections between process entity pairs and relations, which can cause error accumulation. The pipeline-based methods achieved excellent results in process entity recognition or process relation classification. For example, Yao et al. (2020) proposed a symmetric lightweight deep multi-network collaboration (ALBERT-AttBiLSTM-CRF) based entity recognition for manufacturing texts. Kumar and Starly (2022) utilized bi-directional long short-term memory and conditional random field (BiLSTM+CRF) neural network models to extract information from manufacturing science abstracts. Kong et al. (2021) proposed a hybrid natural language processing (hybrid-NLP) algorithm to extract electrical device entities. Shrivastava et al. (2022) employed a Transformer bi-directional encoder language model to recognize manufacturing entities from raw texts. Jing et al. (2022) proposed a named entity recognition algorithm based on the BERT-Imseq2seq-CRF model aimed at facilitating information extraction and domain knowledge graph construction in the context of hot-rolled steel. Guo et al. (2022) established a knowledge extraction framework based on BERT-BiLSTM-CRF for automatic construction of machining process knowledge graph. Guan et al. (2022) used BiLSTM and CR-CNN models to extract process entity relations in unstructured welding process documents. Gu et al. (2022) first used a self-designed BERT-specific attention-CRF model to identify process entities for assembly process documents. Subsequently, they identified relations between process entities based on the designed rules, which is a typical pipeline method. As in the published works, pipeline-based

methods achieved positive results on a single task. In recent years, researchers have focused on joint extraction methods and end-to-end extraction of assembly process knowledge for constructing APKG.

Joint extraction-based methods. These methods can simultaneously extract process entities and process relations. The advantage is that training features can be shared so that the two subtasks depend on each other, thus maintaining the semantic connections between process entities and relations. The joint extraction can solve the problem of isolating two subtasks from each other in the pipeline-based methods and find the globally optimal model parameters during model training, thereby enhancing overall model performance. The joint extraction methods have achieved excellent results in extracting relational triples from simple texts. However, the extraction of assembly process knowledge triples in the manufacturing domain is still nascent, owing to the complexity of assembly process texts. Joint extraction methods present more advantages over previous methods and have begun to attract the attention of researchers. For example, Huang et al. (2023) introduced a cascading decoder to jointly extract entities and relations from unstructured product design data. P. Liu et al. (2023) proposed a joint knowledge extraction model based on reinforcement learning and labeling schemes for the aviation assembly domain, which had high utility value on aviation assembly process documents. Du et al. (2022) proposed a manufacturing knowledge extraction model that applied a simplified graph convolutional net-

work (GCN) to mine entities and extracted dependencies and semantic features through a multi-head stacked GCN.

Knowledge graph-based process planning. The knowledge graph is an advanced method of knowledge representation with inference and self-learning capabilities. These attributes allow it to adapt to the dynamic and variable properties of the assembly process planning. Guo et al. (2022) performed process planning by using a constructed machining process knowledge graph. Kesri et al. (2021) used NLP techniques to extract an automotive knowledge graph from process text data. Then, test processes are generated based on triple matching and context-based search algorithms. Duan et al. (2021) constructed an ontology model-based knowledge graph for metal cutting processes and used the PageRank algorithm to recommend process planning. Zhou et al. (2022) developed an assembly process knowledge graph and designed a sequential knowledge graph convolutional network model to generate assembly process planning. Wang et al. (2022) used a matching algorithm to construct an intelligent Q&A system based on an impeller CNC machining knowledge graph for generating impeller CNC machining processes.

In conclusion, joint extraction-based methods are more suitable than others for addressing the challenges of APKG modeling. Therefore, the proposed model is a joint extraction model, which end-to-end extracts assembly process knowledge and constructs APKG.

Methodology

The pipeline for assembly process planning based on the APKG is shown in Fig. 1. Firstly, historical assembly process documents are preprocessed to transform semi-structured and unstructured documents into corpora that can be annotated. Then, the assembly process dataset is annotated to provide source data for subsequent assembly process knowledge extraction and APKG modeling. The source documents are sourced from the accumulated historical assembly process data of an electronic equipment enterprise. Subsequently, our GRBE model is trained on the dataset, and the trained model achieves assembly process knowledge extraction and automates the construction of the APKG on the test dataset. Finally, the assembly process planning is implemented based on the APKG using our KGMS method.

Data preprocessing

There is no publicly available assembly process dataset, and the assembly process data for this paper are sourced from the historical industrial electronic equipment assembly process documents of an industrial electronic enterprise, totaling

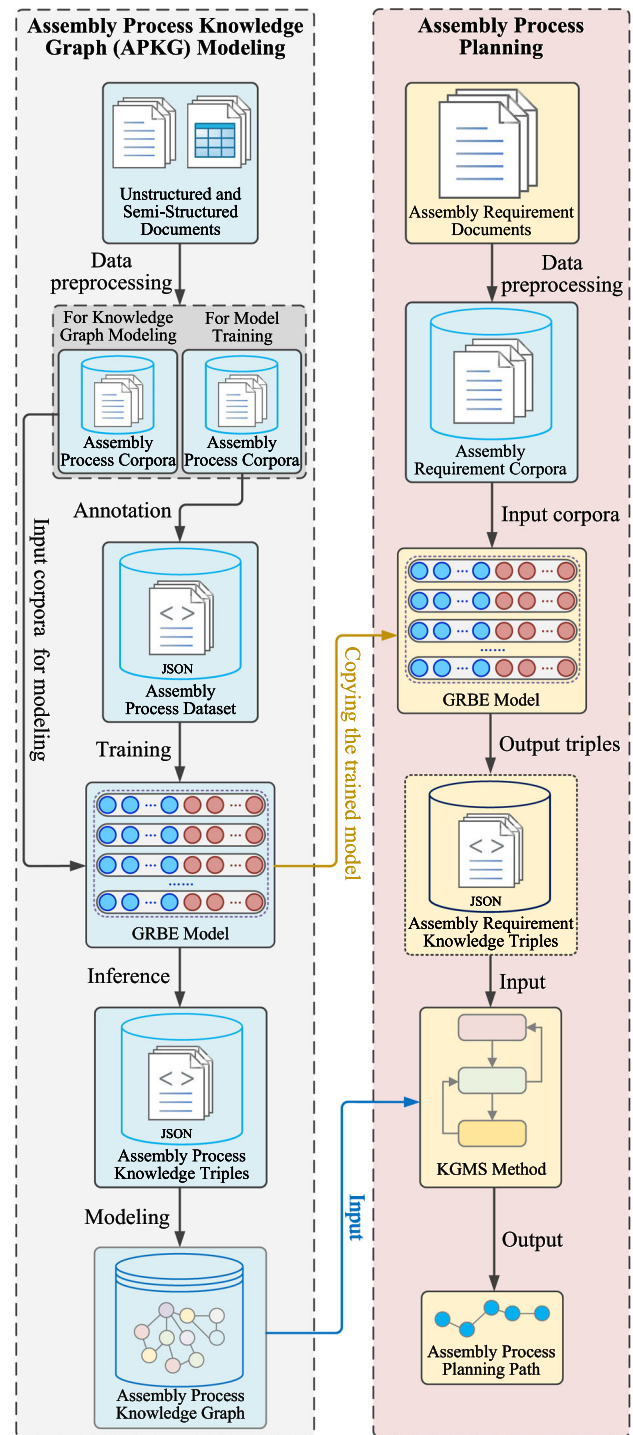


Fig. 1 The pipeline of assembly process planning for industrial electronic equipment based on APKG. The left part shows the APKG modeling procedure, and the right part shows the assembly process planning procedure. The brown arrow indicates the utilization of the trained GRBE model. The blue arrow indicates the utilization of the modeled APKG

320 files. These documents include product design, assembly specification, and assembly process design documents. The documents contain unstructured texts and semi-structured tables. Due to the different data sources and document formats, the documents must be preprocessed. The detailed procedure for document preprocessing and dataset construction is as follows:

- (1) Unstructured documents are examined for meaningful sentences, and the meaningless sentences are separated and removed to form the assembly process corpus that can be labeled.
- (2) Semi-structured tables are extracted automatically according to the table structure rules to form the assembly process corpus, which is merged with the corpus formed in the first step.
- (3) The corpora are analyzed using 5M1E analysis (Gao, He and Wang, 2015) to parse out the assembly entities and assembly relations corresponding to man, machine, material, method, measurement, and environment in the corpora. The thirteen assembly relation categories include perform, target, require, content is, next step, name, place, or, simultaneity, include, is, use, subcategory, and type. The seven assembly entity categories include staff, station, equipment, instrument, tooling, fixture, and material.
- (4) The assembly process corpora are labeled by assembly entity types and assembly relation types.

Knowledge extraction for APKG

In this section, we first present the task definition. Then, we introduce the general framework of our GRBE model. Finally, each module of the model is described in detail.

Task definition

Give an assembly process document $D = \{x_1, x_2, \dots, x_n\}$, where n is the number of tokens. The outputs are the assembly process knowledge triples $T(D) = \{(s, r, o) \mid (s, o) \in E, r \in R\}$ of this document D , where E is the set of assembly entity pairs, R is the set of assembly relations, s is the assembly subject entity, and o is the assembly object entity. The task is jointly extracting all assembly process knowledge triples T in the assembly process document D with an end-to-end method. After completing the assembly process knowledge extraction, the APKG modeling is completed. The visualization of the APKG can be achieved through the graph database. According to the method in this paper, we divide the whole task into four subtasks:

Global relational attention extraction. The assembly relations are semantic connections between assembly entities,

especially for assembly entities contained in long documents. This subtask uses the attention mechanism to mine the implicit assembly relations in the assembly process document D . The output is the assembly relation set $r(D) = \{r_1, r_2, r_3, \dots, r_m\} \subseteq R$ in document D , where m is the number of categories of implicit relations in document D .

High-density Entity Tagging. The density of assembly entities in the assembly process document D is more dense than in normal documents. This subtask tags assembly subject and object entities from the assembly process document D based on the global relational attention mechanism and outputs a coarse-grained assembly entity pair set $e(D)$.

Assembly process knowledge triple combination. This subtask filters coarse-grained entity pair set $e(D)$ to form fine-grained entity pair set $\tilde{e}(D)$ based on the global relational attention mechanism. The output is the assembly process knowledge triple set $T(D)$.

Knowledge graph visualization. The graph database Neo4j (2024) stores and visualizes the assembly process knowledge triple set $T(D)$ and generates the nodes and edges of the APKG.

Model framework

The framework of our GRBE model consists of a pre-trained model encoder, a global relational attention module (GRAM), a subject-to-object tagging module (SOTM), an object-to-subject tagging module (OSTM), and a global relation-based alignment module (GR-based AM), as shown in Fig. 2. The pre-trained model encoder encodes word vector embeddings of the input documents to provide shared features for subsequent modules. The SOTM and the OSTM tag assembly entity subjects and objects in opposite directions. The GR-based AM outputs the assembly process knowledge triples.

Pre-trained model encoder

Given an assembly process document $D = \{x_1, x_2, \dots, x_n\}$, we use the pre-trained model as an encoder to obtain word vector embeddings for each token, as shown in Eq. (1).

$$\begin{aligned} H_e &= \{h_1, h_2, h_3, \dots, h_n\} \\ &= \text{Pre-trained: Model}(\{x_1, x_2, x_3, \dots, x_n\}) \end{aligned} \quad (1)$$

where h_n is the word vector embedding of x_n , and H_e is the word vector embedding of the assembly process document D . The pre-trained model is unsupervised or weakly supervised training on massive corpora to learn contextually relevant word vector embeddings and can be applied to various downstream tasks of natural language processing.

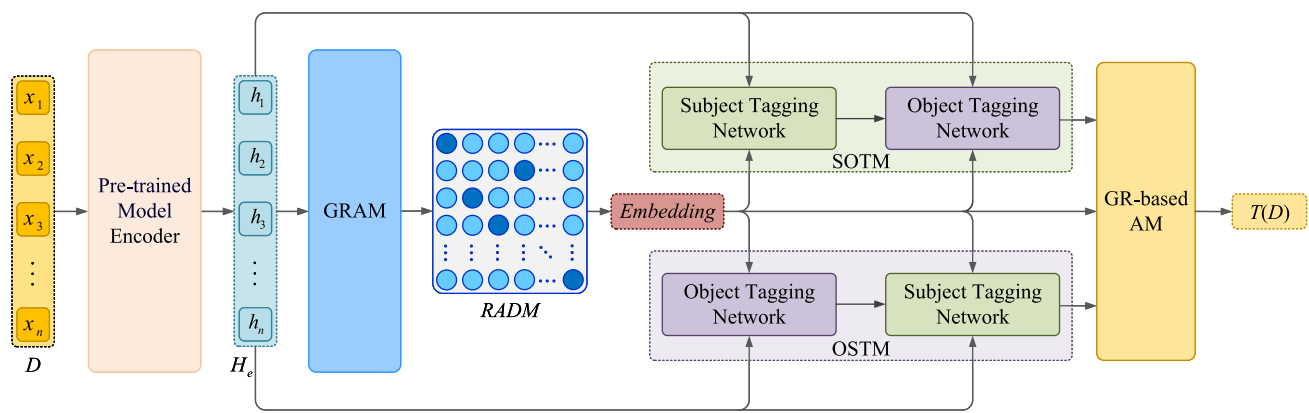


Fig. 2 The architecture of our GRBE model, including a pre-trained model encoder, a global relational attention module (GRAM), a subject-to-object tagging module

(OSTM), and a global relational-based alignment module (GR-based AM). *RADM* is a relation attention distribution matrix

We use the BERT pre-trained model (Devlin, Chang, Lee and Toutanova, 2019; Liu et al., 2019) as an encoder for the model. BERT consists of stacks of identical transformer modules that allow representing the words with contextual semantics.

Global relational attention module

Assembly relations are the semantic connections of assembly entities. Extracting global relations in the assembly process documents can provide prior information for tagging the assembly entities and also enhance the features of the assembly entities. On the other hand, mining the prior global relations can reduce redundant assembly relations. Hence, we propose the GRAM to mine the prior global relations, as shown in Fig. 3. This module extracts the assembly relations from the global perspective to form a relation attention distribution matrix (*RADM*). This module uses four convolution kernels of different sizes to extract relational features from H_e . Convolution kernels of different sizes can extract relational features of different granularity sizes due to different receptive fields. The calculation procedure is shown in Eq. (2).

$$\begin{aligned} H_{conv1} &= \text{ReLU}(W_{r1} * H_e + b_{r1}) \\ H_{conv2} &= \text{ReLU}(W_{r2} * H_e + b_{r2}) \\ H_{conv3} &= \text{ReLU}(W_{r3} * H_e + b_{r3}) \\ H_{conv4} &= \text{ReLU}(W_{r4} * H_e + b_{r4}) \end{aligned} \quad (2)$$

where $*$ denotes the convolution operation, and ReLU is the activation function. $W_{r1} \in \mathbb{R}^{3 \times d}$, $W_{r2} \in \mathbb{R}^{4 \times d}$, $W_{r3} \in \mathbb{R}^{5 \times d}$ and $W_{r4} \in \mathbb{R}^{5 \times d}$ are the training weights, $b(\cdot)$ is a bias vector. $H_{conv1} \in \mathbb{R}^{1 \times n}$, $H_{conv2} \in \mathbb{R}^{1 \times n}$, $H_{conv3} \in \mathbb{R}^{1 \times n}$ and

$H_{conv4} \in \mathbb{R}^{1 \times n}$ are the initial assembly relation feature vectors.

After obtaining the initial assembly relation feature vectors with different granularity sizes, these features are fused by concatenation. The calculation procedure is shown in Eq. (3).

$$H_{conv} = \text{Concat}(H_{conv1}, H_{conv2}, H_{conv3}, H_{conv4}) \quad (3)$$

where Concat is the concatenation function and $H_{conv} \in \mathbb{R}^{4n \times 1}$ is the initial assembly relation feature vector after concatenation.

The *RADM* is generated through fully connected layers and activation functions, as shown in Eq. (4). We introduce the residual network structure (He, Zhang, Ren and Sun, 2016) to address the degradation problem that may be suffered when the model is too deep.

$$\begin{aligned} P_r &= \sigma(\text{ReLU}(W_r * H_{conv} + b_r) \\ &\quad + \text{ReLU}(W_{r5} * H_e + b_{r5})) \end{aligned} \quad (4)$$

where W_r and W_{r5} are training weights, b_r and b_{r5} are bias vectors, and σ denotes the sigmoid function. Set a threshold λ for the distribution value in the P_r , if the distribution value exceeds the threshold λ tag it as 1 or tag it as 0 otherwise. Because an assembly process document contains multiple assembly relations, the tagged P_r would have multiple tags 1. The relation distribution vector is expanded into the *RADM* $\in \mathbb{R}^{m \times k}$ based on the number and position of tags 1. k is the number of tags 1, which is the number of categories of potential assembly relations in the assembly process document mined by the GRAM.

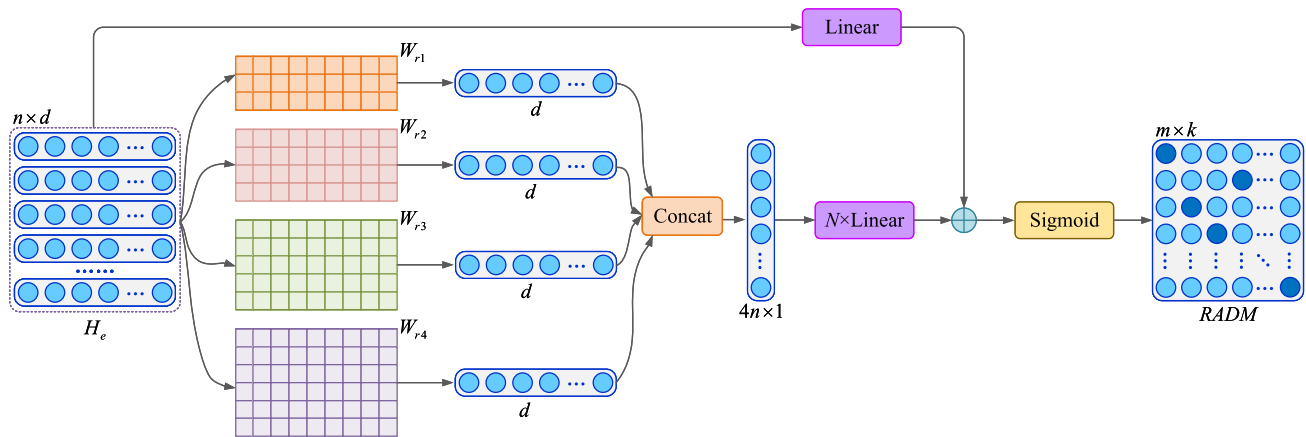


Fig. 3 The architecture of GRAM. The inputs are word vector embeddings of tokens, and the output is the *RADM*

Subject-to-object tagging module

The assembly process documents have high assembly entity density and weak semantic correlations between assembly entities. Therefore, we tag the assembly entities from two opposite directions to enhance the extraction of high-density assembly entities. The features of global assembly relations fused in the tagging of assembly entities enhance the semantic correlation between assembly entities. The first tagging module is the SOTM, as shown in Fig. 4. Combining global relational features to tag the assembly entity subject. Each token of the assembly process document is tagged using the BIO (B-begin, I-inside, O-outside) tagging scheme (Lample, Ballesteros, Subramanian, Kawakami and Dyer, 2016), which clearly distinguishes the head, middle, and tail of entities in the document. The computation procedure is shown in Eq. (5).

$$P_s^{i,k} = \text{Softmax} (W_s [h_i; r_k] + b_s) \quad (5)$$

where $h_i \in \mathbb{R}^{1 \times d}$ is the encoding vector of the i -th token, and $r_k \in \mathbb{R}^{1 \times d}$ is the word vector embedding of the k -th relation in the *RADM*. $P_s^{i,k}$ denotes the $\{B, I, O\}$ probability distribution that the i -th token is the assembly entity object. $W_s \in \mathbb{R}^{2d \times 3}$ is a training weight and b_s is a bias vector.

The features of each token of the tagged assembly entity subject are average pooled, preserving all the semantic features, which can be calculated by Eq. (6).

$$g_s^j = \text{Avgpool} (h_s^{j,B}, \dots, h_s^{j,I}) \quad (6)$$

where $h_s^{j,B}, \dots, h_s^{j,I}$ denotes the feature vector of the j -th assembly entity subject, $g_s^j \in \mathbb{R}^{1 \times d}$ is its feature vector after average pooling, and Avgpool denotes the average pooling operation.

Each token fuses the average feature of the assembly entity subject and the feature of the global relation to tag the assembly entity object, and the computation procedure is shown in Eq. (7).

$$P_o^{i,k} = \text{Softmax} (W_o [g_s^j; r_k; h_i] + b_o) \quad (7)$$

where $P_o^{i,k}$ denotes the $\{B, I, O\}$ probability distribution that the i -th token is the assembly entity object. $W_o \in \mathbb{R}^{3d \times 3}$ is a training weight and b_o is a bias vector.

Object-to-subject tagging module

The second tagging module is the OSTM, which is similar in structure to the STOM, but the tagging direction is opposite. Combining global relational features to tag the assembly entity object as shown in Eq. (8).

$$P_o^{i,k} = \text{Softmax} (W_o [r_k; h_i] + b_o) \quad (8)$$

where $h_i \in \mathbb{R}^{1 \times d}$ is the encoding vector of the i th token, and $r_k \in \mathbb{R}^{1 \times d}$ is the word vector embedding of the k -th relation in the *RADM*. $P_o^{i,k}$ denotes the $\{B, I, O\}$ probability distribution that the i -th token is the assembly entity object. $W_o \in \mathbb{R}^{2d \times 3}$ is a training weight and b_o is a bias vector.

The features of each token of the tagged assembly entity object are average pooled, preserving all the semantic features, which can be calculated by Eq. (9).

$$g_o^j = \text{Avgpool} (h_o^{j,B}, \dots, h_o^{j,I}) \quad (9)$$

where $h_o^{j,B}, \dots, h_o^{j,I}$ denotes the feature vector of the j -th assembly entity subject, $g_o^j \in \mathbb{R}^{1 \times d}$ is its feature vector after average pooling, and Avgpool denotes the average pooling operation.

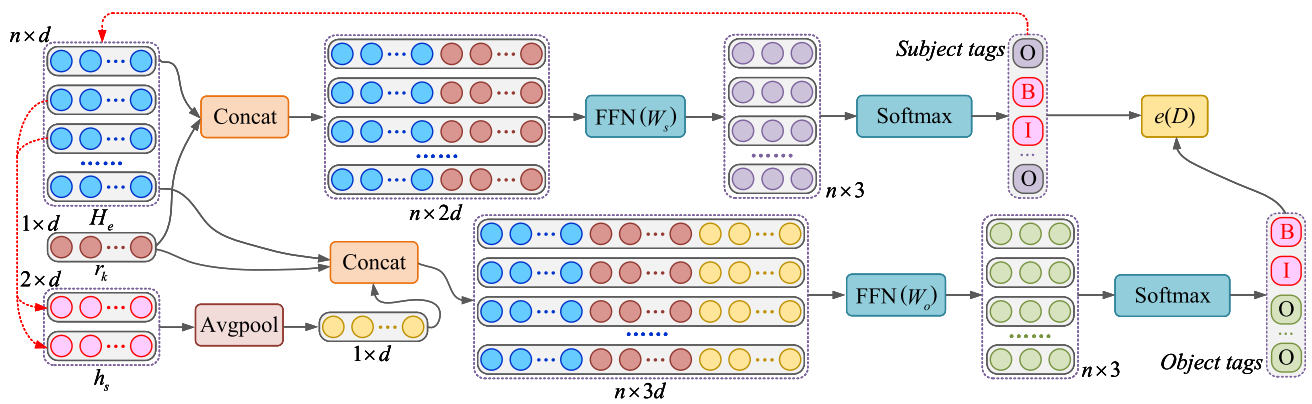


Fig. 4 The architecture of SOTM

Each token fuses the average feature of the assembly entity subject and the feature of the global relation to tag the assembly entity object, and the computation procedure is shown in Eq. (10).

$$P_s^{i,k} = \text{Softmax} \left(W_s \left[h_i; r_k; g_o^j \right] + b_s \right) \quad (10)$$

where $P_s^{i,k}$ denotes the $\{B, I, O\}$ probability distribution that the i -th token is the assembly entity object. $W_s \in \mathbb{R}^{3d \times 3}$ is a training weight and b_s is a bias vector.

Global relation based alignment module

The alignment module filters the coarse-grained assembly entity pair set $e(D)$ tagged with SOTM and OSTM to form fine-grained assembly entity pair set $\tilde{e}(D)$. It combines assembly process knowledge triples $T(D)$ based on global assembly relations. This module eliminates redundant and incorrect assembly entity pairs and combines them with correct assembly relations. The relations between the coarse-grained assembly entity pairs (s_u, o_v) are first judged, and the calculation procedure is shown in Eq. (11).

$$P_{rel}^l = \text{Softmax} \left(W_{rel} \left[h_s^{u,B}; h_o^{v,B} \right] + b_{rel} \right) \quad (11)$$

where $P_{rel}^l \in \mathbb{R}^{1 \times |R|}$ denotes the distribution vector of relation categories for the l -th assembly entity pair and $|R|$ is the number of categories for all assembly relations. $h_s^{u,B} \in \mathbb{R}^{1 \times d}$ and $h_o^{v,B} \in \mathbb{R}^{1 \times d}$ are the encoding vectors of entity pairs (s_u, o_v) corresponding to the start token of the subject and object, respectively. $W_{rel} \in \mathbb{R}^{2d \times |R|}$ is a training weight and b_{rel} is a bias vector.

The Jaccard index is used to judge the similarity between the prior global assembly relation set R_g mined by the GRAM and the aligned assembly relation set R_a formed by Eq. (11), and the calculation procedure is shown in Eq. (12). If $J > 0.9$, output the assembly process knowledge triples based on

the assembly relations in R_a . Otherwise, add the assembly entity pairs to the next GRAM training. Jaccard index is not desired in the APKG modeling stage, and the modeling stage is shown in Algorithm 1.

$$J(R_g, R_a) = \frac{|R_g \cap R_a|}{|R_g| + |R_a| - |R_g \cap R_a|} \quad (12)$$

Algorithm 1 APKG modeling

Input: The assembly process document $D = \{x_1, x_2, \dots, x_n\}$;

Output: $APKG$;

- 1: Word embeddings H_e for D are obtained using the PLM;
- 2: Initialize prior global assembly relation set R_g , and alignment assembly relation set R_a ;
- 3: Initialize coarse-grained assembly entity pair set $e(D)$ and fine-grained assembly entity pair set $\tilde{e}(D)$;
- 4: Initialize assembly process knowledge triple set $T(D)$;
- 5: Input H_e to GRAM, output $RADM \in \mathbb{R}^{m \times k}$;
- 6: Convert the $RADM$ to $r_1, r_2, \dots, r_k$;
- 7: Add the $r_1, r_2, \dots, r_k$ to R_g ;
- 8: **for** each $r \in R_g$ **do**
- 9: Embedding(r);
- 10: Input H_e and Embedding(r) to SOTM, output (s, o) ;
- 11: Add (s, o) to $e(D)$;
- 12: Input H_e and Embedding(r) to OSTM, output (s, o) ;
- 13: Add (s, o) to $e(D)$;
- 14: Input H_e and Embedding(r) to GR-based AM, output $(\tilde{s}, \tilde{o})$ and R_a ;
- 15: Add $(\tilde{s}, \tilde{o})$ to $\tilde{e}(D)$;
- 16: Add $(\tilde{s}, r_a, \tilde{o})$ to $T(D)$, $r_a \in R_a$;
- 17: **end for**
- 18: $APKG \leftarrow \text{Neo4j}(T(D))$;
- 19: **return** $APKG$

Joint training strategy and loss function

We use a joint training strategy to share the feature parameters of the pre-trained model encoder, and the feature parameters between the modules in the training stage can be constrained.

The loss function for each module of the model is defined as follows.

The loss function of the GRAM is based on the binary cross-entropy loss function, as shown in Eq. (13).

$$\mathcal{L}_r = -\frac{1}{|R|} \sum_{i=1}^{|R|} (\hat{y}_i \log P_{r,i} + (1 - \hat{y}_i) \log (1 - P_{r,i})) \quad (13)$$

where $|R|$ is the number of categories of all assembly relations, and $\hat{y}_i \in \{0, 1\}$ is the truth value label.

The loss function of the SOTM is based on the cross-entropy loss function, as shown in Eq. (14).

$$\mathcal{L}_{so} = -\frac{1}{n} \sum_{k=1}^{|r|} \sum_{i=1}^n \sum_{\hat{t}_s \in \{B, I, O\}} \log P_s^{i,k} (t_s = \hat{t}_s) \quad (14)$$

where $|r|$ is the number of potential relation categories in *RADM*, and n is the number of tokens.

The loss function of the OSTM is based on the cross-entropy loss function, as shown in Eq. (15).

$$\mathcal{L}_{os} = -\frac{1}{n} \sum_{k=1}^{|r|} \sum_{i=1}^n \sum_{\hat{t}_o \in \{B, I, O\}} \log P_o^{i,k} (t_o = \hat{t}_o) \quad (15)$$

where $|r|$ is the number of potential relation categories in *RADM*, and n is the number of tokens.

The loss function of the GR-based AM is based on the cross-entropy loss function, as shown in Eq. (16).

$$\mathcal{L}_{rel} = -\frac{1}{|L|} \sum_{l=1}^{|L|} \sum_{\hat{t}_{rel} \in R} \log P_{rel}^l (t_{rel} = \hat{t}_{rel}) \quad (16)$$

where $|L|$ is the number of coarse-grained assembly entity pairs.

Finally, the total loss function for the joint training of the model is composed of the loss functions of the four modules, as shown in Eq. (17).

$$\mathcal{L}_{total} = \mathcal{L}_r + \mathcal{L}_{so} + \mathcal{L}_{os} + \mathcal{L}_{rel} \quad (17)$$

KGMS method for assembly process planning

Assembly process planning is performed on the APKG using our proposed KGMS method, which mainly consists of process node matching and process path searching, as shown in Algorithm 2.

Process node matching

The process nodes in the APKG contain multiple assembly process knowledge triples. First, the process requirement

Algorithm 2 KGMS method for Assembly Process Planning

Input: Process requirement document R , $APKG$

Output: Process planning path $Path$;

```

1: Initialize process knowledge triple set  $T(R) \leftarrow \emptyset$ , process node set  $G \leftarrow \emptyset$ , Process planning path set  $Path \leftarrow \emptyset$ ;
2: Extract the assembly process knowledge triples in  $R$ ,  $T(R) \leftarrow GRBE(R)$ ;
3: Calculate the average word vector embedding  $H(R)$  of  $T(R)$  based on Eq. (18);
4: for  $V \in APKG$  do
5:   Calculate the average word vector embedding  $H(V)$  of  $T(V)$  based on Eq. (19);
6:   Calculate the similarity  $CS$  between  $H(R)$  and  $H(V)$  based on Eq. (20);
7:   if  $CS > 0.8$  then
8:      $G \leftarrow V$ ;
9:   end if
10: end for
11: Initialize a temp set  $G' \leftarrow \emptyset$ ;
12:  $G' \leftarrow G$ ;
13: repeat
14:   Randomly select seed node  $SN \leftarrow V', V' \in G'$ ;
15:    $Path \leftarrow SN$ ;
16:   for  $V \in (G' - Path)$  do
17:     if Discover the nearest node  $V'$  of  $V$  then
18:        $Path \leftarrow V''$ ,
19:        $V'' \in APKG$  includes all the nodes on the path from  $V'$  to  $V$ ;
20:        $Path \leftarrow V$ ;
21:        $SN \leftarrow V$ ;
22:     end if
23:   end for
24:    $G' \leftarrow \emptyset$ ;
25:    $G' \leftarrow (G - Path)$ ;
26: until  $G'$  is  $\emptyset$ 
27: return  $Path$ 

```

document containing the elements of man, machine, material, method, measurement, and environment are input into our proposed GRBE model to extract the assembly process knowledge triple set $T(R)$. Then, the average similarity between the $T(R)$ and the assembly process knowledge triple set $T(V)$ in each process node of the APKG is calculated. When the average similarity is over 0.8, the process node V is retained. The process node set $G = \{V_1, V_2, \dots, V_n\}$ is formed after completing the traversal computation. The average similarity is calculated as shown in Eqs. (18)–(20).

$$H(R) = \frac{1}{|T(R)|} \sum_{p=1}^{|T(R)|} \text{Concat}(\text{Emb}(s_p, r_p, o_p)) \quad (18)$$

$$H(V) = \frac{1}{|T(V)|} \sum_{q=1}^{|T(V)|} \text{Concat}(\text{Emb}(s_q, r_q, o_q)) \quad (19)$$

$$CS = \frac{H(R) \cdot H(V)}{\|H(R)\| \times \|H(V)\|} \quad (20)$$

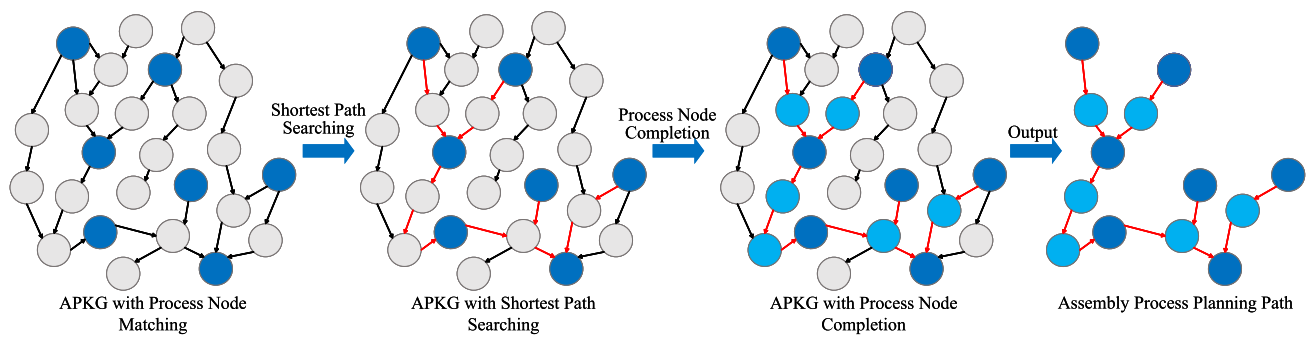


Fig. 5 The assembly process path searching procedure. The dark blue nodes indicate the initial nodes after process node matching, the red arrows indicate the searched process paths, and the light blue nodes indicate the complemented process nodes

where CS denotes cosine similarity, $H(R) \in \mathbb{R}^{1 \times 2304}$ is the average word vector embedding of $T(R)$, and $H(G) \in \mathbb{R}^{1 \times 2304}$ is the average word vector embedding of $T(V)$. $|T(R)|$ and $|T(V)|$ are the lengths of two sets, respectively. Concat denotes the concatenation function, Emb denotes the word vector embedding computation, $\cdot$ is the dot product of vectors, and $\|\cdot\|$ is the vector modulus.

Process path searching

Assembly process paths are represented in the APKG as a topology consisting of multiple process nodes. The shortest path as a search rule on the assembly process knowledge graph is to discover the connecting paths between process nodes in the process node set G and complement the process nodes not in set G . First, match the process nodes in G on the graph. Then, search for the shortest paths between process nodes. Finally, the complete process planning path containing the new process nodes is generated. The details of the algorithm are shown in Algorithm 2 and the searching procedure is shown in Fig. 5.

Experiments

In this section, we conduct experiments to validate the performance of our methods. First, we introduce the experimental settings. Then, we demonstrate the state-of-the-art performance of the GRBE model through comparison experiments with the advanced baselines. Moreover, the model performance is verified through ablation experiments, complex semantic extraction, hyperparameter analysis, and subtask performance experiments. Finally, we construct an industrial electronic equipment APKG and validate the performance of the KGMS method on the test dataset. We also provide an example of assembly process planning for industrial electronic equipment to give researchers a better understanding of our pipeline.

Table 2 Detailed model implementation parameters

Parameters	Values
Token encoding dimension (d)	768
Relation encoding dimension (d)	768
Train batch size	16
Test batch size	16
Valid batch size	16
BERT learning rate	1e-4
Model learning rate	1e-3
Dropout rate	0.3
Epoch	150

Experiment settings

Implementation parameters

Our model is implemented utilizing the PyTorch deep learning framework, and all experiments (model training and model testing) are performed on a workstation that satisfies the experimental requirements. The hardware configuration of the workstation includes Intel i9-13900K@3.00GHz, 128GB RAM, and a single NVIDIA GeForce RTX 4090 GPU. The operating system is Ubuntu 22.04. The BERT pre-trained language model is employed as the encoder in our model. The BERT model (Huggingface, 2024) consists of 12 layers of bi-directional Transformer modules (Vaswani et al., 2017) stacked on top of each other, with an encoding dimension of 768 for each token. The Adam optimization algorithm is chosen as the optimization function of our model. The learning rate for the encoder is set at 1e-4, while the learning rate for the remaining modules is maintained at 1e-3. To mitigate overfitting, each module incorporates dropout layers that randomly deactivate connected units during training, with a dropout rate of 0.3. Additional detailed implementation parameters are presented in Table 2.

Table 3 Statistics of assembly process dataset

	Assembly process dataset			
	Train	Valid	Test	Total
Docs	7975	797	798	9,570
Entities	11,179	3384	3676	18,239
Triples	225,592	24,266	27,538	277,396
Relations	13	13	13	13

Industrial electronic equipment assembly process dataset

According to the data preprocessing procedure in Sect. 3.1, we develop semi-automated annotation software for the industrial electronic equipment assembly process document. The dataset, comprising 320 assembly process files with 9570 documents of about 1277 K words, is divided into train, validation, and test datasets in the ratio of about 10:1:1. The train dataset contains 7975 process documents, 11,179 assembly entities, and 225,592 assembly process knowledge triples. The validation dataset contains 797 process documents, 3384 assembly entities, and 24,266 assembly process knowledge triples. The test dataset contains 798 process documents, 3676 assembly entities, and 27,538 assembly process knowledge triples. There are 13 categories of assembly relations. The statistics of the constructed assembly process dataset are shown in Table 3, and the dataset is stored in the JSON file.

Baselines and evaluation metrics

For evaluating the performance of our model, we use the state-of-the-art models as baselines as follows:

- **DulDec** (Ma, Ren and Zhang, 2021). This model uses an efficient cascading bi-decoder consisting of a text-specific relation decoder and a relation-corresponding entity decoder.
- **PFN** (Yan, Zhang, Fu, Zhang and Wei, 2021). This method proposes a partition filtering network modeling the interaction between entities and relations, ensuring the interdependence between both tasks.
- **SPN** (Sui, Zeng, Chen, Liu and Zhao, 2023). This model treats joint entity and relationship extraction as an ensemble prediction problem, allowing the model to escape the burden of predicting multiple triples.
- **OneRel** (Shang, Huang and Mao, 2022). This method converts the joint extraction into a fine-grained triple classification and solves the joint method suffering from cascading errors and redundant information.
- **PRGC** (Zheng et al., 2021). This model is an end-to-end extraction framework that decomposes relational triple extraction into three subtasks.

- **CasRel** (Wei, Su, Wang, Tian and Chang, 2020). This model first extracts the subjects and then simultaneously extracts the relations and the corresponding objects.
- **ConCasRTE** (Ren et al., 2021). This method is a conditional cascade tagging model similar in structure to CasRel.
- **GRTE** (Ren et al., 2021). This method is a table-filling based relational triple extraction method and incorporates global features.

In this paper, we adopt generic metrics for evaluation, including the standard micro Precision (Prec.), Recall (Rec.), and F1-score (F1) (Junker, Hoch and Dengel, 1999; Pereira, Plastino, Zadrozny and Merschmann, 2018). The calculation of each metric is presented in Eqs. (21)–(23).

$$P_{micro} = \frac{\sum_{i=1}^L TP_i}{\sum_{i=1}^L TP_i + \sum_{i=1}^L FP_i} \quad (21)$$

$$R_{micro} = \frac{\sum_{i=1}^L TP_i}{\sum_{i=1}^L TP_i + \sum_{i=1}^L FN_i} \quad (22)$$

$$F1_{micro} = \frac{2 \times P_{micro} \times R_{micro}}{P_{micro} + R_{micro}} \quad (23)$$

where TP_i (True Positive) denotes the number of i th class positive samples that are correctly predicted. FP_i (False Positive) denotes the number of i th class positive samples that are incorrectly predicted. FN_i (False Positive) denotes the number of i th class negative samples that are incorrectly predicted. $F1_{micro}$ is a composite evaluation indicator that is the harmonic mean of P_{micro} and R_{micro} .

Experimental results and discussions

Main results

For the fairness of the comparison experiments, in addition to the experiments on our constructed assembly process dataset, the experiments are also conducted on the publicly available assembly equipment fault diagnostic dataset. The assembly equipment fault diagnostic dataset is obtained from DataFountain (2024) and consists of 1491 annotated assembly equipment fault diagnostic documents. It is divided into train, validation, and test datasets in the ratio of 10:1:1. The train dataset consists of 1250 documents, the validation dataset consists of 120 documents, and the test dataset consists of 121 documents, where the fault relations contain four categories, including component fault, performance fault, detection tool, and composition.

Table 4 shows the comprehensive experimental results of our model on the assembly process dataset and the fault diagnostic dataset. Our model exhibits superior performance across both datasets, achieving the highest precision, recall,

Table 4 Main results

Models	Assembly process dataset			Fault diagnostic dataset		
	Prec	Rec	F1	Prec	Rec	F1
DulDec (Ma, Ren and Zhang, 2021)	54.1	51.5	52.8	62.1	64.2	63.1
PFN (Yan, Zhang, Fu, Zhang and Wei, 2021)	91.3	88.8	90.1	81.8	80.7	81.3
SPN (Sui, Zeng, Chen, Liu and Zhao, 2023)	–	–	–	62.8	58.7	81.3
OneRel (Shang, Huang and Mao, 2022)	89.6	72.1	79.9	60.8	55.1	57.8
PRGC (Zheng et al., 2021)	91.2	92.2	91.7	62.8	39.5	48.5
CasRel (Wei, Su, Wang, Tian and Chang, 2020)	54.2	64.7	59.0	66.3	19.7	30.4
ConCasRTE (Ren et al., 2021)	93.5	79.9	86.2	81.1	39.3	52.9
GRTE (Ren et al., 2021)	81.3	80.9	83.4	55.4	62.8	59.6
Our (GRBE)	93.6	92.2	92.9	88.7	87.0	87.9

Comparative experimental results of our model on the assembly process dataset and the fault diagnostic dataset, including micro precision (%), recall (%), and F1 (%)

Bold indicates the highest score. '–' indicates no result. Prec. is micro precision, Rec. is micro recall, and F1 is micro F1-score

and F1-score. Our model demonstrates a significant enhancement in recall and F1-score on the assembly process dataset, showing improvements of 12.3% and 6.7%, respectively, compared to ConCasRTE. This demonstrates that our model can extract the correct process knowledge triples more adequately with more balanced performance. While the PRGC model also attains a relatively balanced performance on the assembly process dataset, our model surpasses it by enhancing precision by 2.4% and F1-score by 1.2%. On the fault diagnostic dataset, our model demonstrates improvements across all evaluation metrics in comparison to other models. The comparative analysis of the results indicates that the performance of our model on the fault diagnostic dataset is generally lower than that on the assembly process dataset because the fault diagnostic dataset is much smaller. The assembly process dataset comprises 9570 documents, whereas the fault diagnostic dataset includes only 1250 documents. These results show that our model has advantages in the knowledge triple extraction of long documents, complex semantic assembly relations, and high-density assembly entities.

Ablation studies

In this section, we investigate the effectiveness of each module in our model by performing ablation experiments on the assembly process dataset. The precision, recall, and F1-score of the model are obtained after removing the module to analyze the difference in the experimental results. The results of the ablation experiments are shown in Table 5.

To study the effectiveness of the GRAM, this module is removed. Each token word vector embedding output by the BERT encoder is input to the SOTM and OSTM, which do not fuse the assembly relation word vector embeddings output by the *RADM* in the assembly entity tagging. Furthermore, GR-

Table 5 Results of ablation studies (%)

Models	Prec	Rec	F1
GRBE	93.6	92.2	92.9
-GRAM	84.4	33.1	47.5
-SOTM	93.2	92.1	92.7
-OSTM	93.0	92.1	92.6
-GR-based AM	56.1	87.8	68.4

'–' indicates a removal operation

based AM does not use the prior global assembly relations. As shown in Table 5, the removal of GRAM results in a precision reduction of 9.2% and a more significant recall decrease of 59.1%. The decrease in recall far surpasses the decrease in precision, which proves that the model loses the ability to extract the global assembly process knowledge triples. The experimental results demonstrate that the GRAM can effectively improve the model performance.

To authenticate the effectiveness of the SOTM, the OSTM is employed exclusively as the backbone tagging network for the assembly entities. This identical approach is applied to substantiate the validity of the OSTM. As shown in Table 5, the results of the evaluation metrics are all decreased. It is observed that the results of the evaluation metrics for both modules are closely aligned, suggesting a balanced feature-sharing mechanism between the two modules.

To analyse the effectiveness of the GR-based AM, this module is removed. This means that only the coarse-grained assembly entity pair set $e(D)$ is output. As illustrated in Table 5, there is a notable decrease in precision by 37.5%, a modest decrease in recall by 4.4%, and a significant decrease in F1-score by 24.5%. The significantly lower precision compared to recall demonstrates that the coarse-grained assembly

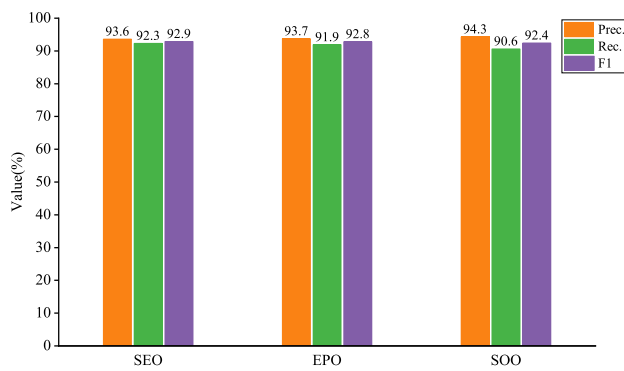


Fig. 6 Extraction results of our model for different complex semantic assembly process knowledge triples on the assembly process test dataset

process knowledge triples are not directly applicable to final results.

Complex assembly semantics and high-density assembly entities

To verify the extraction performance of our model for complex semantic assembly process knowledge triples, we first analyze the forms of complex semantics in the test dataset. Complex semantics have the following formats: single entity overlap (SEO), entity pair overlap (EPO), and subject object overlap (SOO). SEO represents that an assembly entity is involved in multiple assembly process knowledge triples, or multiple assembly relations share an assembly entity. EPO is the same assembly entity pairs with different assembly relations. SOO represents that the assembly subject is the same as the assembly object or partially overlaps. Various complex semantic assembly process knowledge triples are tested in the test dataset, with results presented in Fig. 6. The extraction precision of our model for the three types of complex semantic assembly process knowledge triples is 93.6%, 93.7%, and 93.4%, respectively. The experimental results demonstrate the extraction performance of our model for complex semantic industrial electronic equipment assembly process documents. The difference in extraction precision between the three types of complex semantic documents is within 1%, which proves the stability of our model. The difference between the respective of recall and F1-score are also very small, which also indicates the stability of the model.

The test dataset has numerous assembly process documents for high-density assembly entities. The entity density represents the quantity of assembly process knowledge triples contained in an assembly process document. Figure 7 statistics the quantity of assembly process documents in the assembly process test dataset with entity densities ranging from more than 20 to more than 50. For example, the first bar chart in Fig. 7 indicates that the test dataset con-

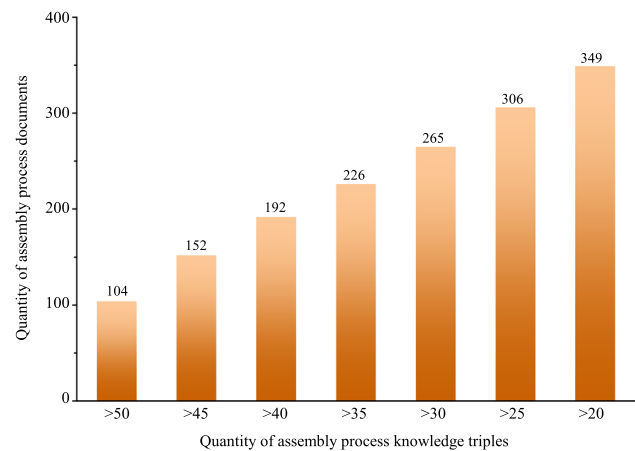


Fig. 7 Statistical results of entity densities in the assembly process test dataset. The entity density represents the quantity of assembly process knowledge triples contained in an assembly process document

tains 104 documents with more than 50 triples. As illustrated in Table 6, our model demonstrates excellent performance across various densities of assembly process knowledge triples exceeding 20. In a scenario exceeding 45, the model achieves a precision of 95.8%, a recall of 95.2%, and an F1-score of 95.5%. In a scenario exceeding 50, the results achieve a precision of 95.7%, a recall of 94.8%, and an F1-score of 95.3%. The difference between the results of the metrics in different scenarios is very small, which proves the stability and robustness of our model for the high-density assembly process knowledge triples.

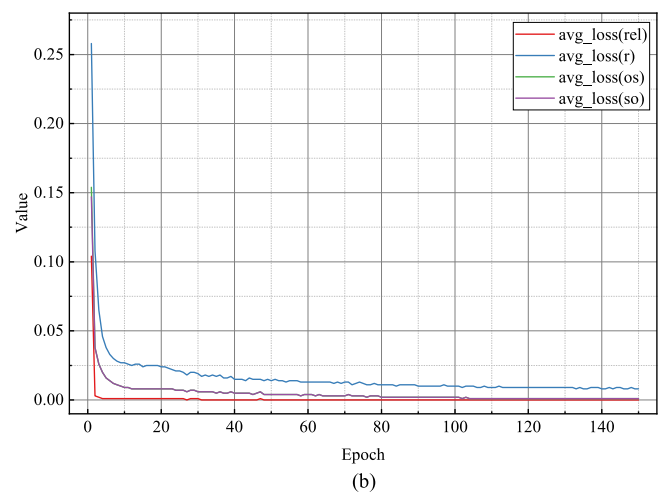
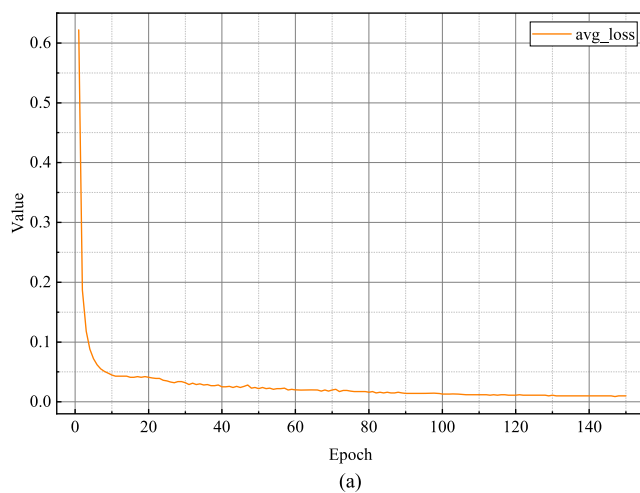
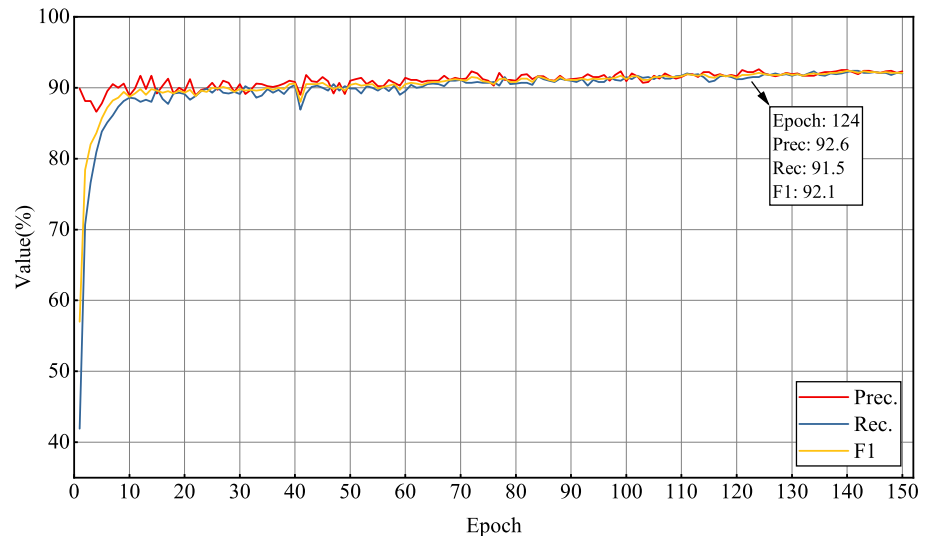
Model convergence

In this section, we study the convergence of the model on the validation dataset. The model performance and loss variations reflect the convergence. As shown in Fig. 8, the precision, recall, and F1-score gradually increase to a steady state within 150 training epochs. During the first 10 training epochs, the values of the three evaluation metrics quickly increased. The precision first decreases and then rapidly increases because the pre-training parameters are not sufficiently adapted to the training dataset. As the model training progresses, the pre-training parameters automatically adapt to the training data, and the precision swiftly increases. At the 30th epoch, almost all evaluation metrics achieve 90%. At the 124th epoch, the precision, recall, and F1-score achieve 92.6%, 91.5%, and 92.1%, respectively. In the subsequent epochs, the three evaluation metrics increase slightly, approaching a stable state.

The average loss variations during the model training stage can better reflect the convergence of the model. As shown in Fig. 9a, the average loss of the model decreases rapidly during the first 10 training epochs, from approximately 0.62–0.04. The average loss decreases slowly and smoothly over

Table 6 Extraction results of our model for different entity density process documents (%)

Entity Densities	> 50	> 45	> 40	> 35	> 30	> 25	> 20
Prec	95.7	95.8	94.5	94.6	94.5	94.5	94.7
Rec	94.8	95.2	94.7	94.8	94.2	93.9	93.7
F1	95.3	95.5	94.6	94.7	94.3	94.2	94.2

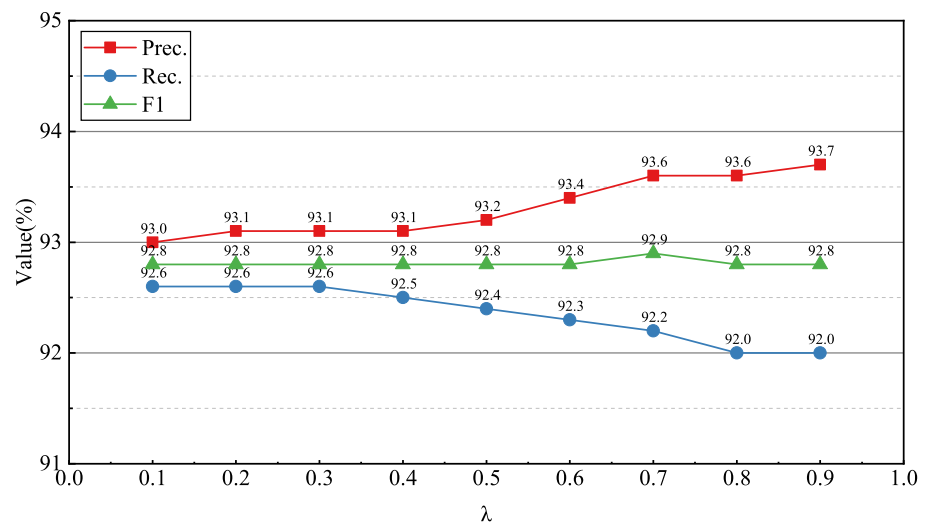
Fig. 8 Variations of precision, recall, and F1-score on the validation dataset during model training**Fig. 9** Variation curves of average loss during model training: **a** shows the variation curves of model average loss, **b** shows the variation curves of sub-module average losses. Where avg_loss(rel) is the aver-

age loss for GRAM, avg_loss(r) is the average loss for GR-based AM, avg_loss(os) is the average loss for OSTM, and avg_loss(so) is the average loss for SOTM

the subsequent 140 epochs, with a loss value of 0.01 at the end. As shown in Fig. 9b, the trend of the average loss for each sub-module is consistent with the model. The average loss of the GRAM module reduces from 0.15 to 0.001 during the first 5 training epochs, which is faster than the other sub-modules. It proves that the GRAM converges to the optimal training parameters before the other sub-modules. It also demonstrates the reasonableness of the model framework, as the GRAM needs to provide global relational attention to

the other sub-modules, which must be trained in advance. The average loss and convergence of SOTM and OSTM are almost the same, which verifies the balanced sharing of training parameters and features between the two modules during the training stage.

Fig. 10 Performance variations of the model for different thresholds on the assembly process test dataset



Model parameter analysis

To focus on the effect of the most important hyperparameter λ threshold in our model for the model performance, we set different thresholds to verify the model performance. The thresholds are varied from 0.1 to 0.9, with the corresponding validation results depicted in Fig. 10. The precision fluctuating between 93.0–93.7% indicates a proportional relationship between precision and threshold variation. As the threshold increases, the precision of the model correspondingly increases. The recall fluctuating between 92.6% and 92.0% demonstrates an inverse proportionality to the threshold variation. The increased threshold allows the model to extract more assembly process knowledge triples, but this comes at a higher error rate. Across all three metrics, the variations are confined to less than 1%. The experimental outcomes reveal that the model exhibits substantial tolerance towards threshold variations, confirming its stability and robustness.

Results of subtasks

To evaluate the extraction efficiency of our model for various subtasks, we decompose the assembly process knowledge triple (AS, AR, AO) into its constituent elements. AS is the assembly subject, AR is the assembly relation, AO is the assembly object, and AE is the assembly entity. Table 7 illustrates that the model achieves the best extraction performance for AR, with 3.5% increase in precision, 3.3% increase in recall, and 3.4% increase in F1-score compared to the extraction results of (AS, AR, AO). This result demonstrates the effectiveness of GRAM for the extraction of assembly relations. The 2.9% increase in precision, 2.6% increase in recall, and 2.8% increase in F1-score for the AE demonstrate that our model can also be applied to named entity recognition

for assembly entities. The difference between AS and AO is negligible, which validates the effectiveness of SOTM and OSTM. The performance of the model on (AS, AR) and (AR, AO) is lower than that on AE, AS, and AO, which validates that incorrect global relations may result in performance degradation.

Case studies of assembly process planning

The proposed model is utilized to construct the APKG on the electronic equipment assembly process test dataset. The APKG is a semantic network structure where assembly entities are the nodes and assembly relations are the edges. The model extracts two types of assembly process knowledge triples from the test dataset: annotated in-scope triples and other newly discovered triples. The extracted assembly process knowledge triples are stored in the graph database Neo4j (2024) and visualized after entity disambiguation and alignment. The constructed APKG has 3256 nodes and 7778 edges, including 13 categories of assembly relations and 7 categories of assembly entities. Owing to the extensive size of the graph, a partial visualization is presented in Fig. 11.

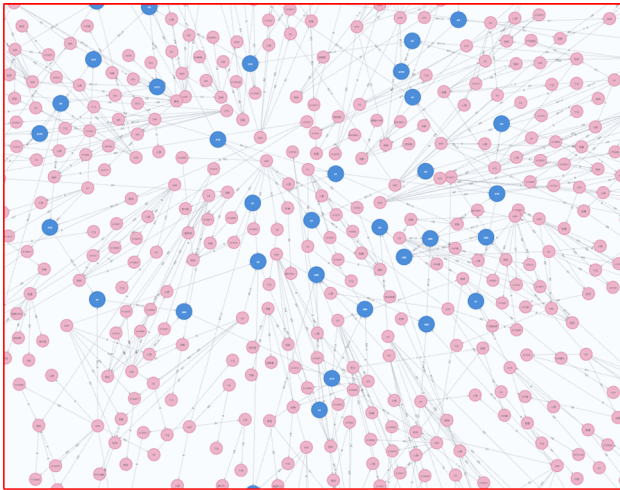
The APKG is not just for the management of assembly process knowledge but also for the intelligent process planning in response to actual production requirements. Traditional CAPP systems (Yusof and Latif, 2014) merely simplify the process design flow but cannot adapt to dynamic variations in the production environment and the diversity of process types. The KG has the ability for inference and self-learning (Guo et al., 2020). Therefore, the APKG can adapt to the dynamics and variability in the equipment assembly procedure (Xiao, Zheng, Shi, Du and Hong, 2023).

To validate the performance of our KGMS method for process planning on the industrial electronic equipment APKG, we conduct experiments with randomly selected 13 unseen

Table 7 Performance of our model on different subtasks (%)

Elements	Prec	Rec	F1	D-Prec	D-Rec	D-F1
Assembly Entity (AE)	96.5	94.8	95.7	+ 2.9	+ 2.6	+ 2.8
Assembly Subject (AS)	96.2	92.8	94.5	+ 2.6	+ 0.6	+ 1.6
Assembly Relation (AR)	97.1	95.5	96.3	+ 3.5	+ 3.3	+ 3.4
Assembly Object (AO)	96.5	94.8	95.6	+ 2.9	+ 2.6	+ 2.7
(AS, AR)	95.7	92.7	94.2	+ 2.1	+ 0.5	+ 1.3
(AR, AO)	94.8	93.2	94.0	+ 1.2	+ 1.0	+ 1.1
(AS, AR, AO)	93.6	92.2	92.9			

D-Prec. denotes the variation of precision (%), D-Rec. denotes the variation of recall (%), and D-F1 denotes the variation of F1-score (%)

**Fig. 11** A partial example of the constructed industrial electronic equipment APKG

process documents from the dataset as test data. We use the Jaccard index as the accuracy metric for the generated assembly process planning paths, and the experimental results are shown in Table 8. The KGMS method achieves the highest accuracy of 96.0%, in which 25 process path nodes are generated, containing all the original 24 process path nodes. The lowest accuracy is 71.2%, in which the ground truth quantity of process path nodes is 60, and 52 nodes are predicted, showing a decrease in accuracy with a longer process path. The average accuracy is 85.0%, which proves the effectiveness of the KGMS method in combination with APKG in generating process planning paths. It is worth noting that although the average accuracy of 85.0% is not perfect, to our knowledge, the KGMS method is the first quantitative method that evaluates the performance of an automated process planning method other than just giving an example demonstration. Therefore, it can be further optimized as a baseline for the process planning field.

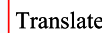
Table 8 Experimental results of the KGMS method

No.	Quantity of process path nodes			Acc	Avg_Acc
	Ground truth	Generated by KGMS	Equal to ground truth		
1	4	5	4	80.0	85.0
2	60	65	52	71.2	
3	13	12	11	78.6	
4	55	53	47	77.1	
5	17	20	16	76.2	
6	22	25	22	88.0	
7	24	23	20	74.1	
8	15	16	15	93.8	
9	11	13	11	84.6	
10	27	27	26	92.9	
11	24	25	24	96.0	
12	15	14	14	93.3	
13	17	18	17	94.4	

ACC. (%) denotes accuracy and Avg_ACC. (%) denotes average accuracy

Ground Truth denotes the quantity of process path nodes in the test dataset. Generated by KGMS denotes the quantity of process path nodes generated by the KGMS method. Equal to Ground Truth denotes the quantity of generated process path nodes that are the same as the ground truth process path nodes

(a) A visual example of the generated assembly process planning path.



Process Contents: “ At station ‘F34’ , staff ‘E45’ use a type ‘C695’ equipment to tag content ‘H266’ in material ‘K1593’ as required by design drawing ‘DW1’ .”

(b) Local zoom-in view of process node A153.

Finally, the assembly process knowledge triples are output through GR-based AM. Experiments achieve F1-score of 92.9% on our constructed industrial electronic equipment assembly process dataset. The experiments verify the effectiveness of the submodules, and our model achieves excellent performance on the subtasks. Additionally, we construct the industrial electronic equipment APKG using our GRBE model and propose the KGMS method to support process planning on the APKG. These experimental results illustrate the effectiveness of the proposed pipeline for assembly process planning. In future work, we will extend the proposed pipeline to other equipment assembly fields, such as automotive or aerospace manufacturing. We hope that the proposed method in this paper can contribute innovative ideas for assembly process planning in equipment manufacturing.

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Author Contributions Youzi Xiao: methodology, investigation, validation, writing—original draft. Shuai Zheng: writing—original draft, resources, conceptualization. Jiewu Leng: formal analysis, writing—review and editing. Ruibo Gao: visualization, investigation, writing—review and editing. Zihao Fu: visualization, investigation, writing—

review and editing. Jun Hong: writing—review and editing, conceptualization.

Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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